NASA SPACE APPS · PLANET X AND SPHEREx

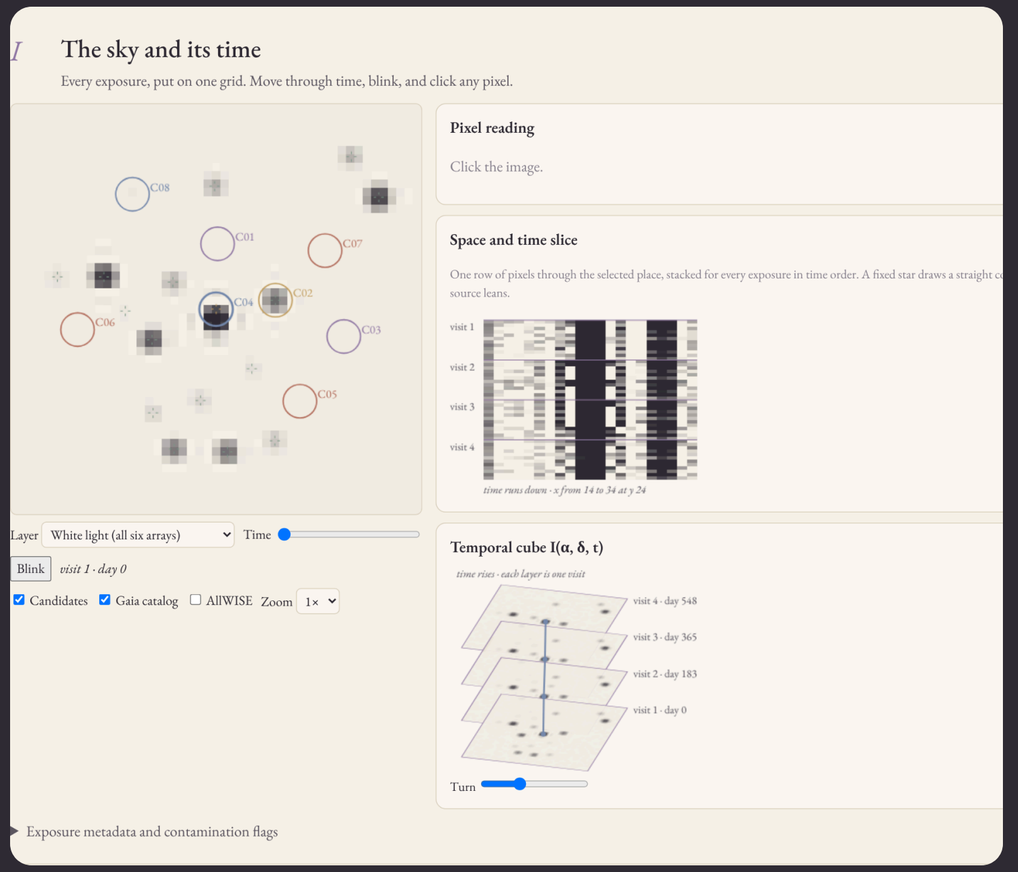
Nakshatra CHRONOS

Same sky. Same color.
Then ask what changed.

An observatory that finds change in the SPHEREx infrared sky, and then tries to prove itself wrong.

3 detectors | 5 hypotheses | 400 test fields | 61 tests

For watchers of the changing sky, like Radhagobinda Chandra of Jessore, and for anyone who wants to check a claim.



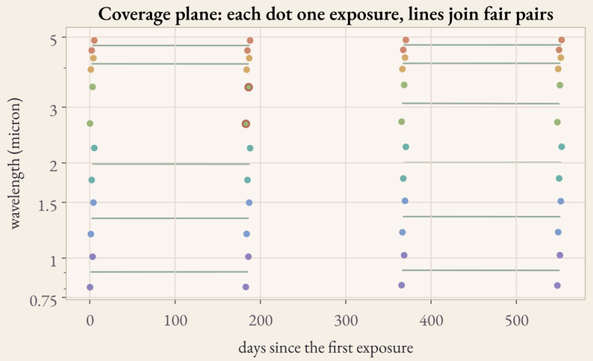
pritidipadas.github.io/nakshatra-chronus · github.com/PRITIDIPADAS/nakshatra-chronus

You cannot blink SPHEREx

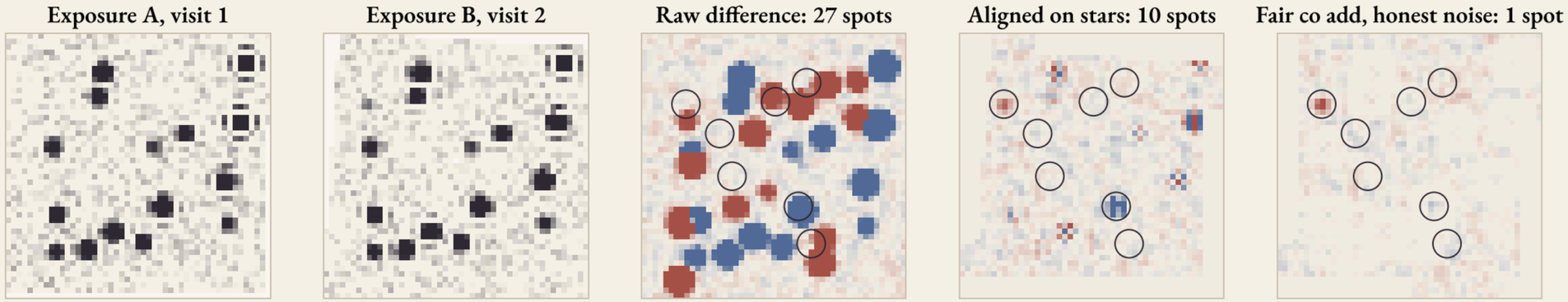
A linear variable filter gives every pixel its own color, 0.75 to 5 micron in 102 channels. Surveys 3 and 4 sit half a channel away.

A steady star can look 1 to 2 percent brighter from color alone. For a faint mover like a distant planet, false change can hide it or fake it.

So CHRONOS compares only **fair pairs**: same array, color within a third of a channel, one release.



The difference microscope: array 2, the same pair with more corrections



Three ways of seeing. Their disagreement is a measurement.

Fair pairs → P · D · A → Candidates → Five hypotheses → Tests that try to break it → Evidence report

P · catalog

Sources in each visit, linked, with motion and brightness floors measured on stars.

D · difference

Fair co adds of two visits; the resampling error is measured, not guessed.

A · anomaly

Isolation forest, local outlier factor and robust covariance rank each pixel in time.

34%
of changes seen by one method are real

77%
seen by two

91%
seen by all three

5%
of changes seen only by the anomaly model are real; the rest is the camera

512 candidates from 109 practice fields the engine never saw. Agreement is not proof: a crosstalk copy fooled all three, and the instrument tests caught it.

Every candidate must defend itself

Five hypotheses, and the right to say unknown

Moving, variable, transient, artefact, or no change. Calibrated.

The self questioning machine

Star drift, one visit left out, the array behind the dichroic, crosstalk.

The counterfactual universe

The whole analysis again under eleven other reasonable choices.

A debate of three analysts

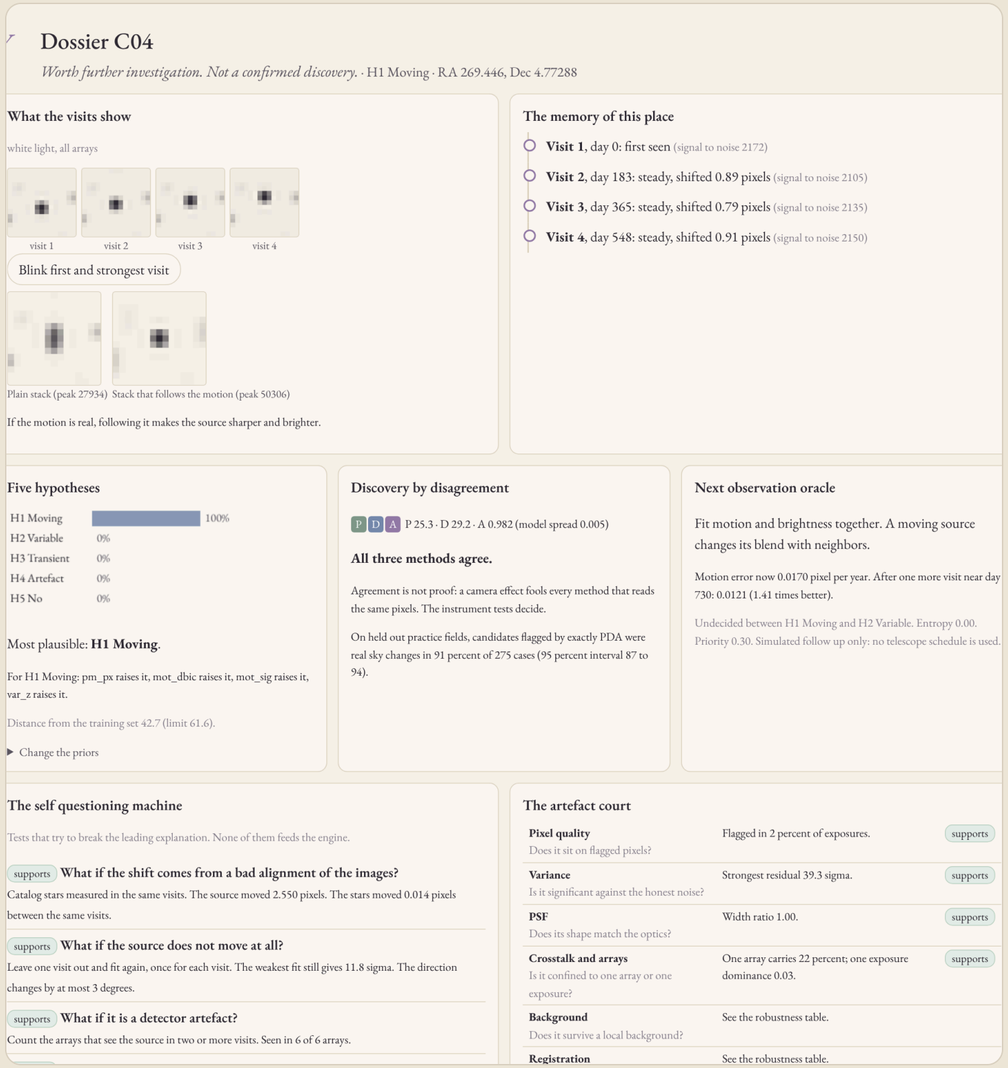
Motion, photometry, instrument. Disagreement stays on the record.

All three methods agreed, and they were wrong together: a crosstalk copy of a bright star (array 1 only)



C05: flagged by PDA
Verdict: Most consistent with an instrumental artefact.

- The partner does not see it.
- Yes. A bright star sits 128 columns to the right on the same row in f3_e1_d4_s1.
- 86 percent in one array.



Evidence, not adjectives

98.8%
right when the engine decides, 109 unseen fields

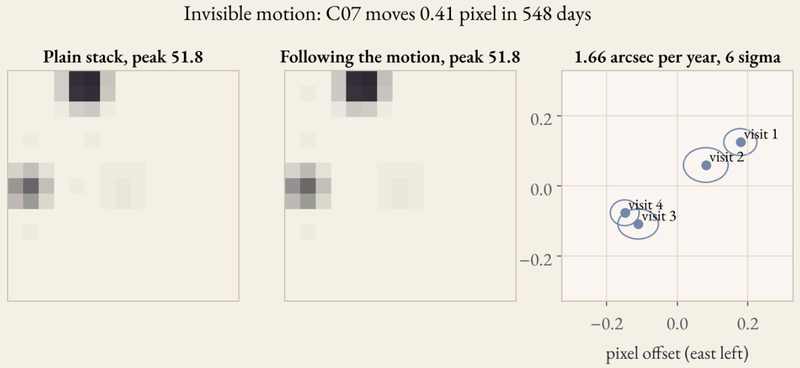
0.007
calibration error: a stated 80 percent is right 80 percent of the time

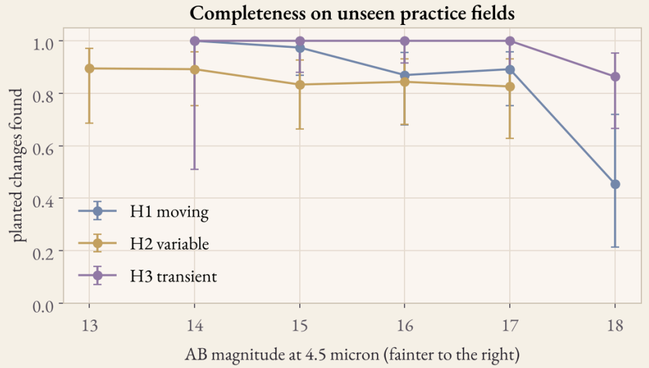
117 of 117
glitches, hot pixels and crosstalk copies rejected

0 of 840
catalog stars with no motion called moving

0.375
false candidates per empty field

6 sigma
motion of 0.4 pixel, invisible to the eye





Practice sky with a hidden truth. On real QR3 files only 24 percent of planted sources were found, and the practice engine said unknown, not a false claim.

From one blink to Planet X, honestly

Rung 1

Theoretical prediction

Orbits suggest an unseen mass.

Rung 2 · CHRONOS reaches here

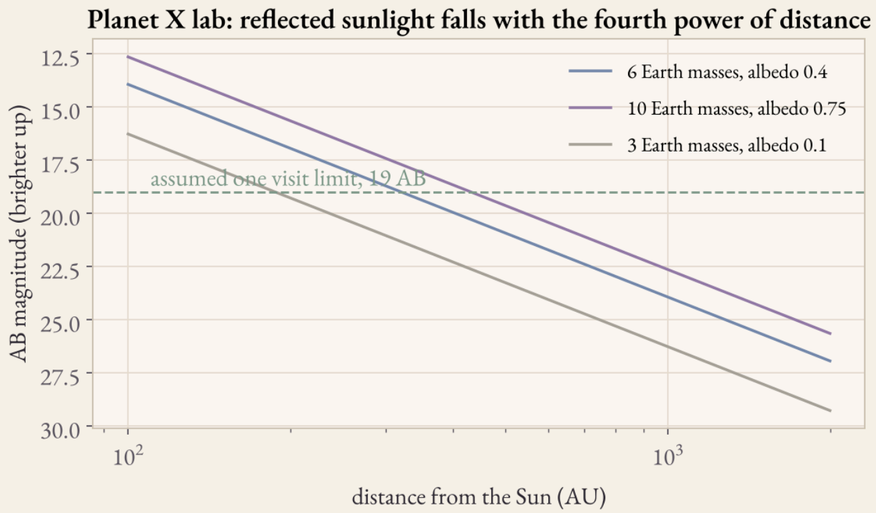
Observational candidate

A source that moves the right way.

Rung 3

Supported detection

Artefacts excluded, false alarms measured.



SPHEREx scientists

A screen that ranks candidates and shows why.

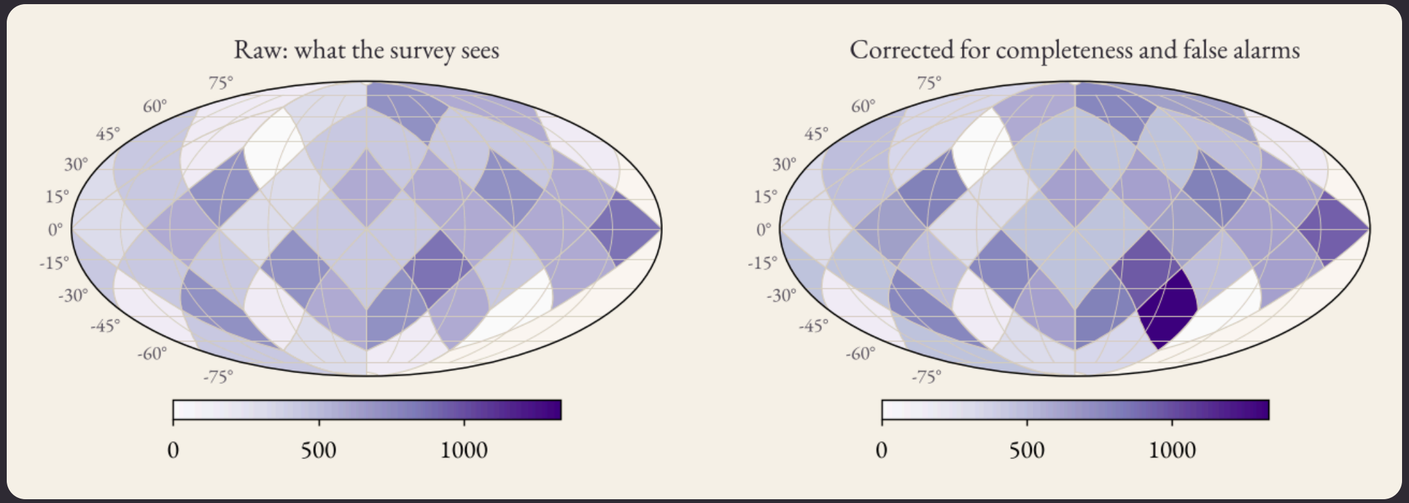
Students and citizens

A citizen lab, and the story in Bangla.

Everyone

A sealed record that anyone can reproduce.

Built and tested. Then the real sky.



Whole sky map

Deep tiles: 45 percent found raw. Corrected: 106 percent of the truth.

Citizen study

Same 16 cases for all, 4 blind repeats, Fleiss kappa.

Rubin cross match

Rubin alerts through ALeRCE, known bodies from JPL.

Real QR3 injection

120 planted in real files: 29 found. Retrained engine: 12 of 17 right.

Next

- Retrain on real noise and deep fields.
- Whole sky tiles from real QR3 files.
- Run the study with a class of students.
- Add NEO Surveyor after its launch.

Open the live observatory.
Apache 2.0 · 73 tests
Docker · 4 notebooks

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